

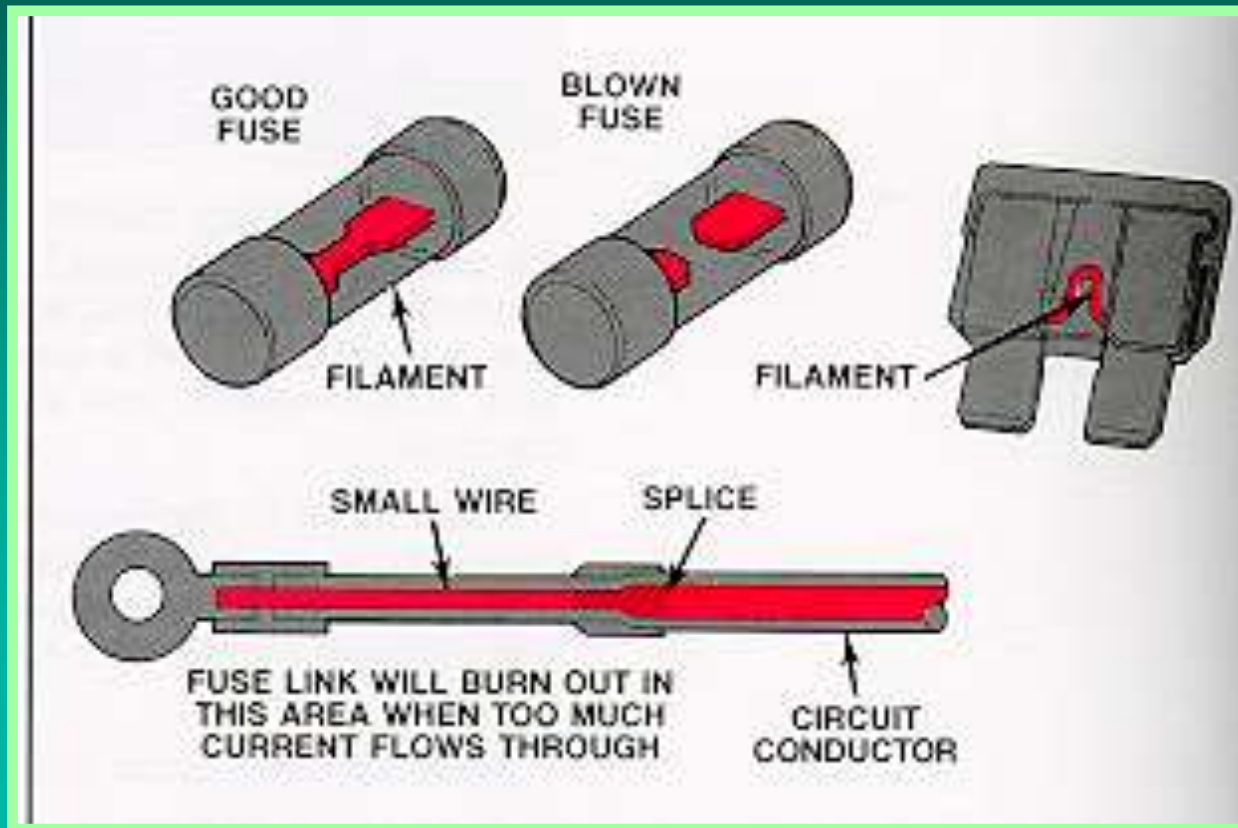
ELECTRICAL COMPONENTS

Circuit Protection



Circuit Protection

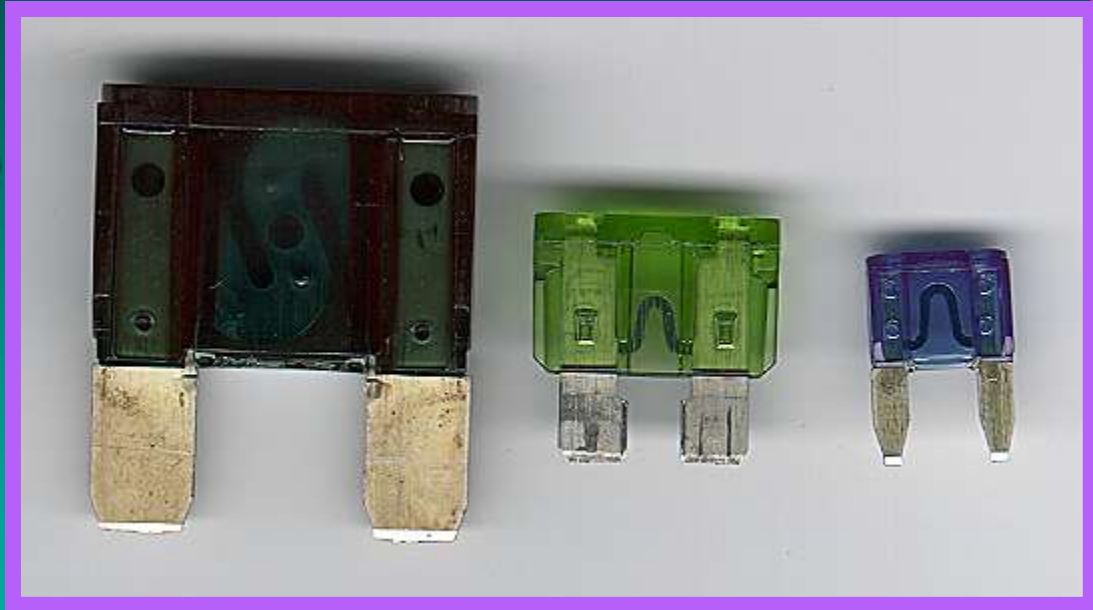
■ Types of Fused Protection



*How to
pull them*

Fuses

- Auto-fuse
(blade type
and Color
coded)
- Mini-fuse
- Maxi-fuse



A. TEST HOLES
B. REMOVAL

Maxi-fuses

- Combination blade / cartridge
- Protects main circuits
- Safer than fusible link
- Cover fewer circuits than a fusible link
- Often in Power Distribution box

Ceramic and Glass

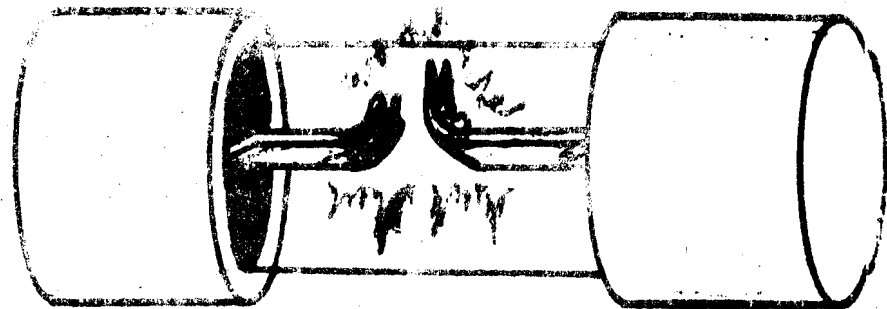
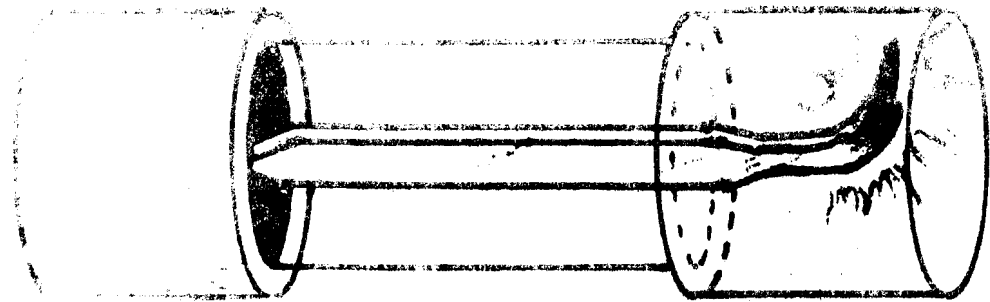


- Rated by current failure level
- Three letter code for type and size
- Glass style replaces ceramic
- Caution in pulling

Two reasons for blowing

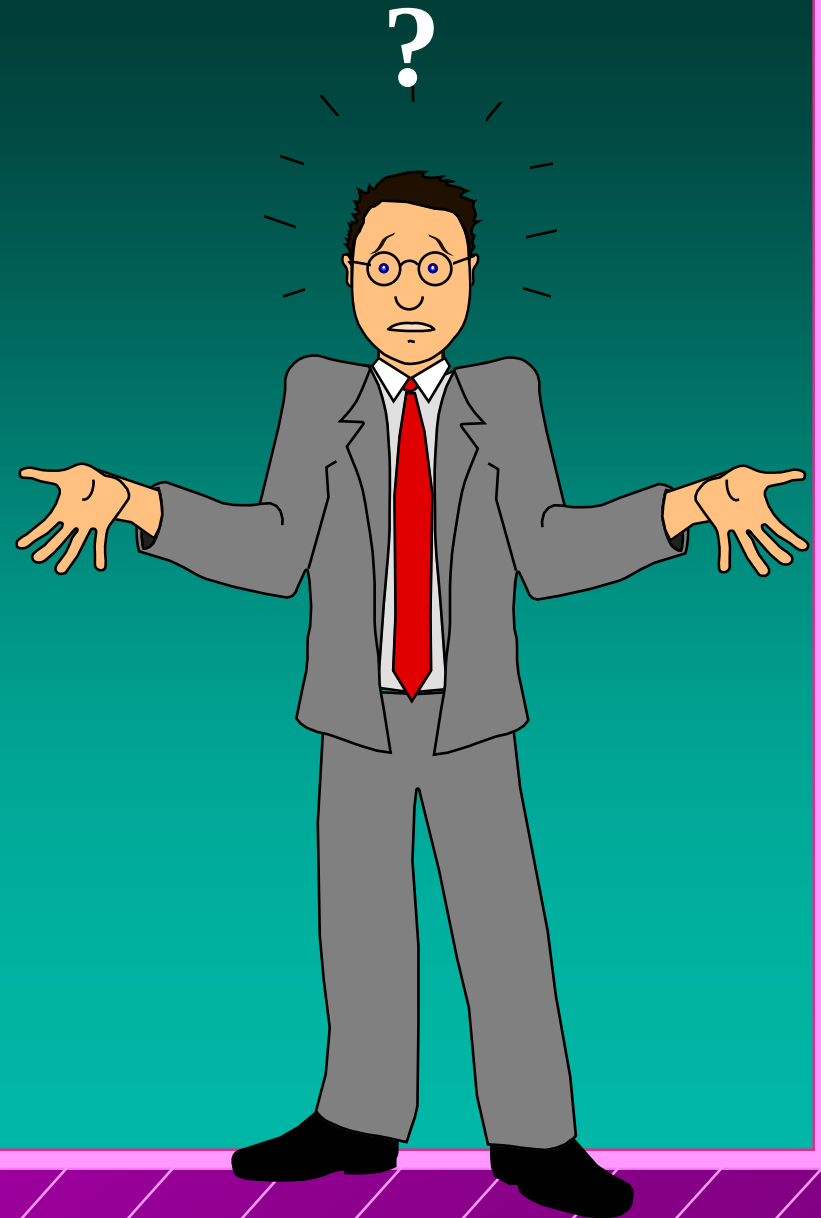
Where do they blow?

Why?



New Circuits

- What size of fuse should I install?
- Use Watts law.
Watts divided by volts



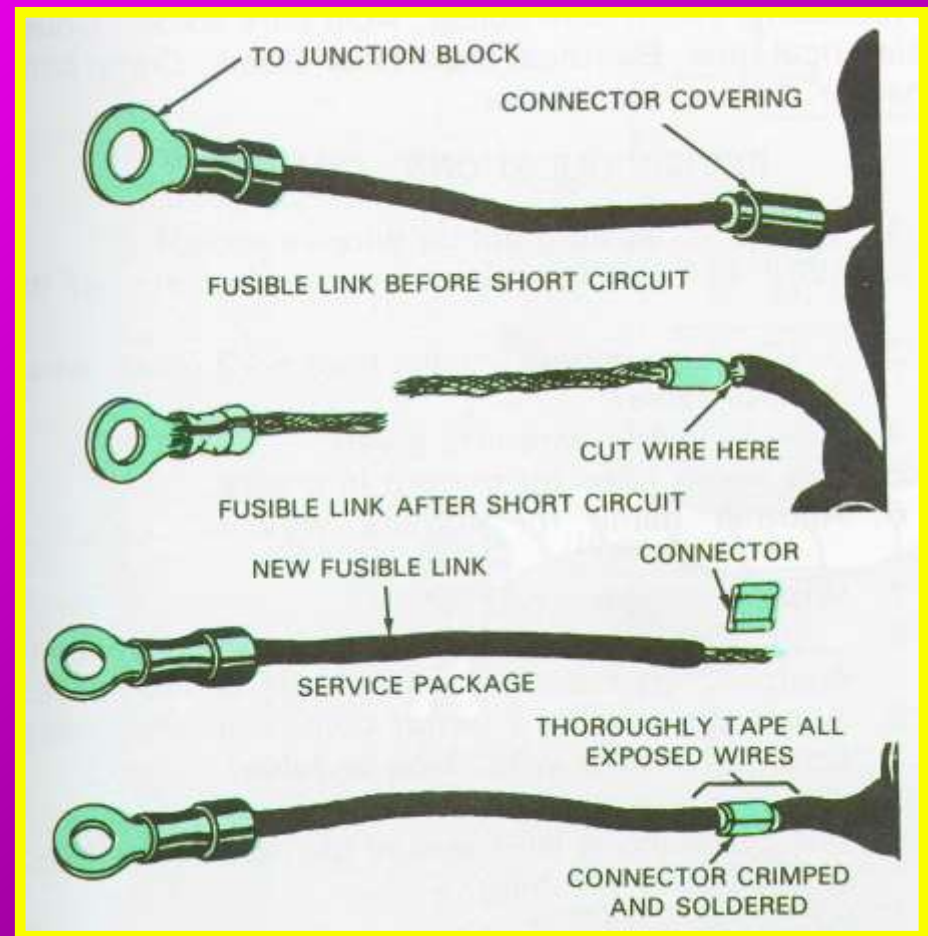
FUSIBLE LINKS



- Lighter gauge wire than main conductor
- Covered with special insulation
- Protect Main circuits
- Usually under hood

Fusible Links Repair

- Location
- Circuits protected
- Insulation
 - Visual checks
- Installing a new link
 - 4 wire sizes smaller (4 numbers larger)
 - Soldering



Fuses



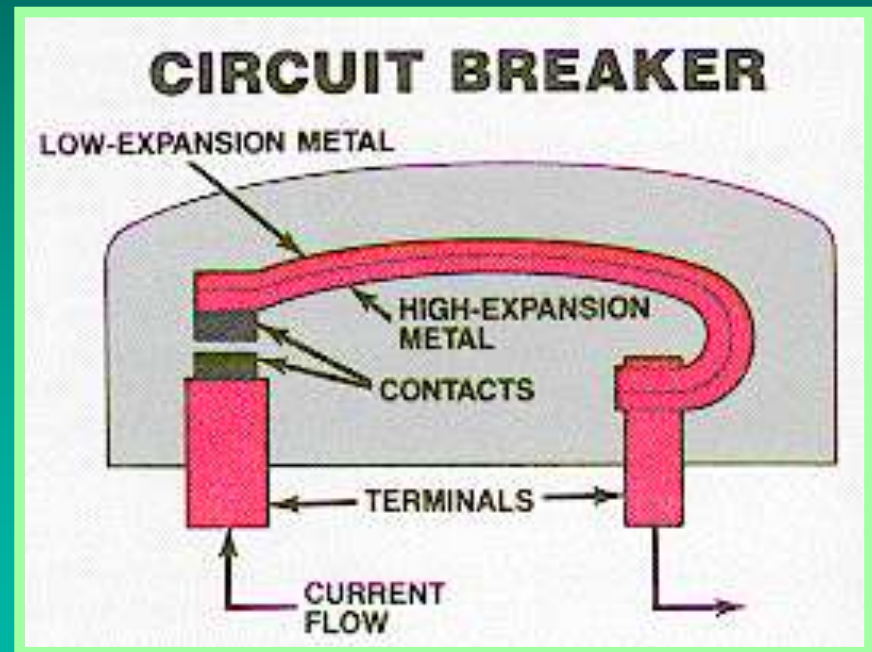
Male Pal fuses

Female Pal fuses

Bolt on Pal Fuses

Circuit Breaker

- Why circuit breakers
- Styles
- Deterioration

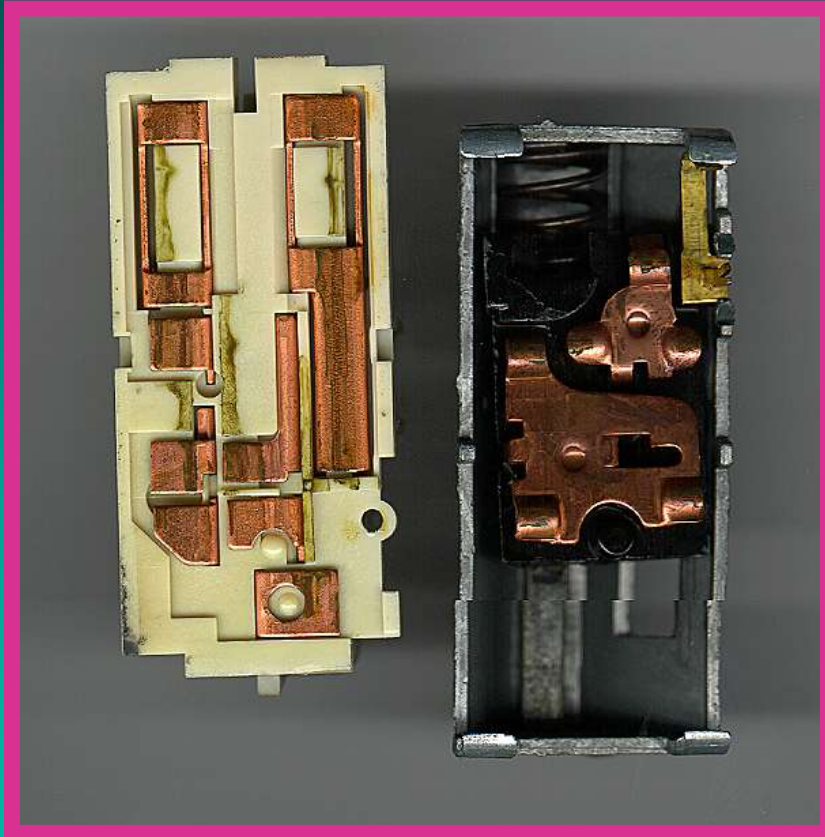


TESTING CIRCUIT PROTECTION DEVICES

- Must inspect closely
- Type of failure determines cause
- Best to use DVOM
- Do not overload circuit by installing too large of fuse or tin foil
- Connections must be tight
- Do not use un-fused jumper wire

Electrical Components

Switches



- Controls electrical current (N.O. or N.C.)
- Single-pole, single throw (SPST)
- Momentary Switch
- Single-pole, double throw switch (SPDT)
- Ganged Switch

Relays

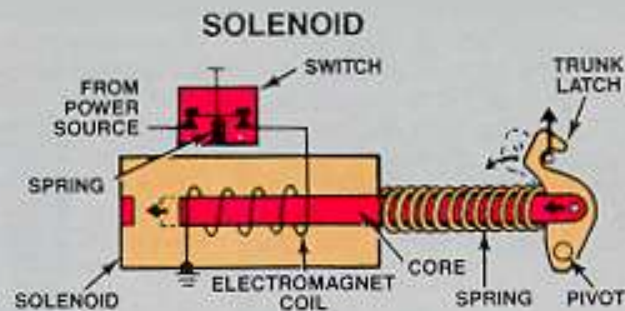
- Electromagnetic Switches (Relays)
- Two Circuits
 - Control Circuit
 - Load Circuit
- Magnetic field operates contacts
- Late model relays are universal



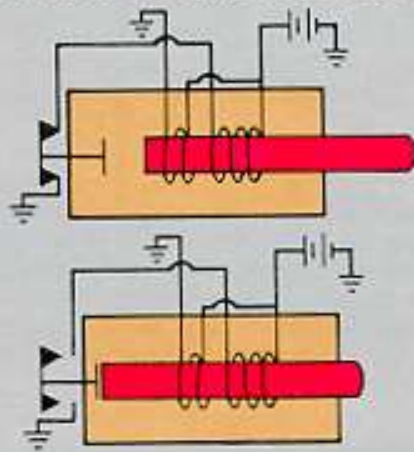
TESTING RELAYS

- Can use several methods to test
- Must Check both circuits
- Be careful using test light if relay is operated by computer
- Can bench test if needed
- Some relays have schematic on them
- Must be correct resistance

SOLENOIDS

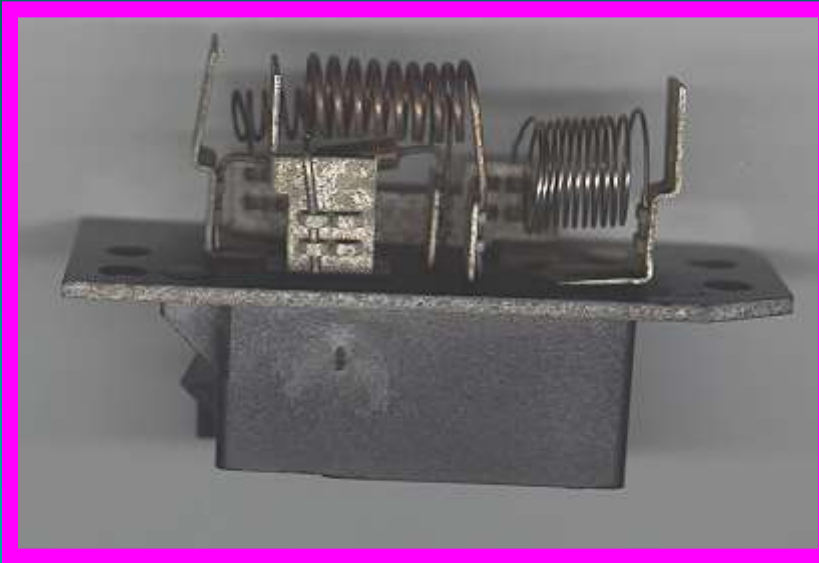


PULL-IN COIL AND HOLDING COIL



- Electromagnetic device with a iron core
- Does mechanical work
- Core is moveable and does work
- Can test with DVOM

STEPPED RESISTORS



- Usually used to control fan motor speeds
- Resistance is changed by control of switch
- Controls current to change speeds
- Thermal fuse

Variable Resistors

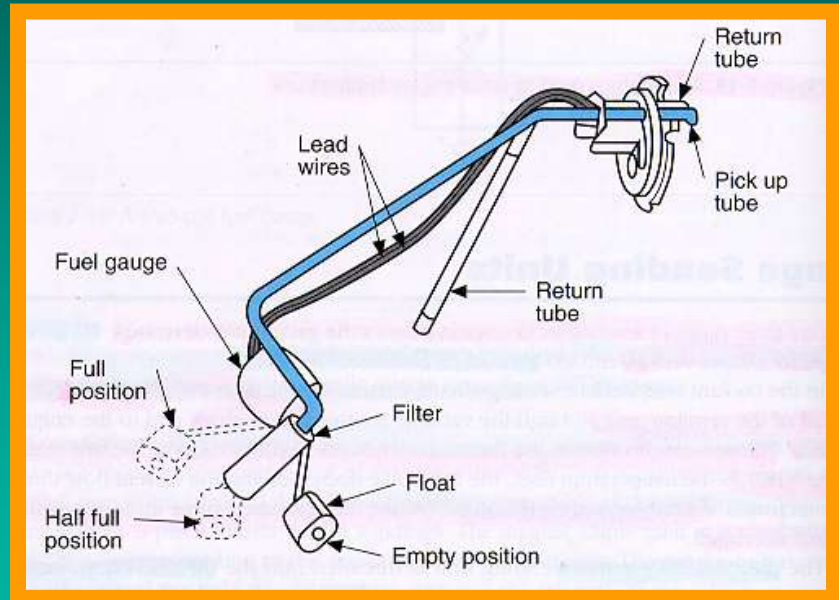
■ Rheostats

- Two terminals
- Higher current

■ Potentiometer

- Three terminals
- Lower current

■ Many uses for variable types



The three types of circuit defects are:

- Shorts
- Grounds
- Opens
 - poor connections

TESTING FOR CIRCUIT DEFECTS

- DVOM IS BEST TO USE!
- Must know circuit operation before can diagnose problem
- Must know how to use equipment and which equipment to use.

**Now, did he say,
“Amps, Volts or
Ohms”?**

